

Aditya, you're 2.7
years
biologically younger
than your calendar.

Lean Metabolically Excellent • Atopic overlay • Circadian-shifted

PHENOAGE

22.3

vs chronological 25.0

RULES FIRED

5

1 Tier-1 • 1 Tier-2 • 3 Tier-3

MARKERS ANALYZED

78

69 from lab • 9 derived

Genetic longevity baseline: exceptional. Four grandparents past 85 — top-decile signal.
This report is about sharpening, not fixing.

Vitamin D severely deficient at 15 ng/mL

This sits below the Tier-1 threshold of 20 ng/mL and warrants a physician-supervised loading protocol (typically 60,000 IU/week × 8–12 weeks, with K2 and magnesium co-supplementation). It is also the **single most important driver** of three of your four sub-optimal findings — eosinophilia, relative-low T, and borderline RDW. Fix this one number, and three others will move.

Also recommended (parallel track): Stool ova-and-parasite (O&P) panel within 2 weeks given recent travel to Bihar + Thailand + eosinophils 9.4%. This is cheap, definitive, and clinically appropriate.

WHAT THIS REPORT SAYS

Five findings worth your attention

- 1. Vitamin D = 15 ng/mL (severely deficient).** This is the headline finding. Physician-supervised loading dose recommended. Will cascade into three other improvements.
- 2. Eosinophils 9.4% + seasonal rhinitis + Bihar/Thailand travel.** Atopic phenotype is clinically confirmed, AND parasitic infection cannot be ruled out without stool O&P testing. Get the test.
- 3. Total testosterone 502 ng/dL.** Above the clinical low-T threshold (400) but below Tessera-optimal (600–900) for your age. Will likely rise to 600–800 with Vit D restoration + earlier dinner + Zone-2 ramp.
- 4. AM cortisol 20 µg/dL.** Not chronic stress (your PSS-4 is 4, in the low band). This is a circadian-misalignment signal — late dinner (11pm–12am) shifts your whole curve. Move dinner to 8–9pm.
- 5. Everything else is excellent.** Lipids, glucose, insulin, liver, kidney, thyroid — top-decile for your age. Stay the course on what's working.

YOUR PHENOTYPE

Lean Metabolically Excellent

Peer, future-focused, optimization framing. You are already in top-decile metabolic shape for your age — every blood-chemistry chassis is humming. The work is at the edges: one severe nutrient deficiency (Vit D), one atopic + post-travel inflammatory signal needing workup, one circadian shift to make. This is sharpening, not fixing.

WHY WE READ IT THIS WAY

- BMI 24.8 (in 18-25, lean-athletic) marker optimal
- Waist 76 cm — waist:height 0.41 (well below 0.5 risk threshold)
- HOMA-IR 0.74 (excellent; optimal <1.0)
- HbA1c 4.8% (optimal; <5.3)
- Fasting insulin 3.80 μ U/mL (optimal; <6)
- Lipids: TC 146, LDL 86, HDL 49, TG 56, ApoB 70 — every marker optimal or near-optimal
- Liver: ALT 17, AST 23, GGT 11, ALP 90, Albumin 5.00 — every
- Kidney: eGFR 105, Creatinine 1.02 — optimal
- Thyroid: TSH 2.064, FT3/FT4 in range — euthyroid
- hsCRP 0.90 (in acceptable; near optimal)
- Homocysteine 7.46 (near optimal)
- Strength training 3-4 $\times$ /wk PPL split — solid base
- No metabolic-syndrome features

OVERLAYS

Atopic / Allergic Overlay — Eosinophils 9.4% (>6% threshold), Eosinophils absolute $0.5 \times 10^3/\mu\text{L}$ (top of normal), HPI confirms seasonal rhinitis (severity 4/10), Total IgE 30 — in normal range BUT atopic phenotype clinically confirmed regardless.

CONSIDERED BUT DOWNGRADED

Stress / Cortisol-Dominant — Cortisol AM 20 $\mu\text{g}/\text{dL}$ initially suggested HPA hyperactivation, BUT: PSS-4 = 4 (LOW band), self-stress 4-5/10, PHQ-2 borderline at 3 (mood 7-8 contradicts depression), fatigue 1, brain fog 1, libido active. Cortisol

reading better explained by circadian phase-shift (sample at 8:53am on likely delayed sleep schedule). Re-routed to a new 'R-MET-CIRC-01' Tier 3 rule.

2 root drivers + 1 medical track — not 5 separate problems

You'll see findings across Vitamin D, cortisol, eosinophils, testosterone, RDW, and phosphorus. Rather than treat each as a separate target, we trace them back to a small set of root drivers. **Attack the roots; the leaves resolve.**

Vitamin D deficiency cascade

CONFIDENCE: **HIGH**

The driver: Severe Vit D deficiency (25-OH = 15 ng/mL) — driver of immune dysregulation (Th2 bias amplifies atopy + eosinophilia), reduced calcium/bone homeostasis, reduced steroidogenesis (suppresses T support), reduced mood/sleep modulation.

EXPLAINS THESE MARKERS

- `vitamin_d_25oh` (15) — definitional
- `eosinophils_pct` (9.4) — Vit D regulates Th1/Th2 balance; deficiency biases toward Th2/atopic
- `testosterone_total` (502) — Vit D is steroidogenic cofactor; deficiency suppresses T synthesis
- `rdw_cv` (14.1) — weakly — chronic micronutrient deficiency contributes to RBC heterogeneity

LIFESTYLE SIGNALS CONFIRM

- Indoor work + 30-min commute (limited UVB)
- Hyderabad latitude has adequate UVB year-round, but exposure is operational not climatic
- No current Vit D supplementation

Attack via: Supplements (loading dose), Recovery (AM sunlight), Nutrition (fatty fish + egg yolks)

Circadian misalignment + sleep-architecture compression

CONFIDENCE:
MODERATE-HIGH

The driver: Late dinner (11pm-12am) → delayed digestion → delayed melatonin → late bedtime (~1am) → cortisol peak shifts later → high-normal AM cortisol on standard 8am draw. Deep-sleep window compressed by both late onset and likely interrupted sleep — directly impacts T synthesis (REM/deep-sleep dependent).

EXPLAINS THESE MARKERS

- cortisol_am (20) — phase-shifted peak rather than chronic elevation
- testosterone_total (502) — compressed deep-sleep window reduces T pulse synthesis

LIFESTYLE SIGNALS CONFIRM

- Dinner 11pm-12am (13h eating window actual vs 10h stated)
- Time to fall asleep 20-40 min (mild sleep-onset latency)
- Bedtime inferred ~1am (late phase)
- PSS-4 = 4 (LOW) — disproves chronic-stress interpretation of cortisol
- Mood 7-8 + libido stable — contradicts HPA-fatigue pattern

Attack via: Recovery (anchor bedtime 11pm, dinner 8-9pm, AM sunlight ≤30 min after wake), Supplements (Mg glycinate bedtime)

Atopic Th2-skewed immunity + post-travel parasitic workup

CONFIDENCE:
MODERATE

The driver: Atopic phenotype (seasonal rhinitis + Th2 bias amplified by Vit D deficiency) confirmed clinically. Parallel: recent travel to Bihar (rural India — endemic for helminths/giardia) + Thailand creates a parasitic differential that **MUST** be ruled out before attributing eosinophilia 9.4% solely to atopy.

EXPLAINS THESE MARKERS

- `eosinophils_pct` (9.4) — Th2 bias OR parasitic — needs workup
- `eosinophils_abs` (0.5) — top of normal — same

LIFESTYLE SIGNALS CONFIRM

- Confirmed seasonal rhinitis (severity 4/10)
- Recent travel: Bihar, Thailand, Goa (last 6 months)
- No GI symptoms currently (mild bloating only)
- No diagnosed food allergies

Attack via: Medical (stool O&P testing, allergist consult if rhinitis worsens), Supplements (quercetin, omega-3, Vit D restoration), Recovery (allergen avoidance, HEPA bedroom)

Leaf findings (explained by the roots above):

- `eosinophils` 9.4% — Multi-causal leaf — primary: Th2 atopic + Vit D deficiency; differential: post-travel parasitic. Workup via stool O&P recommended.
- `testosterone` 502 — Leaf of M-02 (Vit D) + circadian root. Don't intervene at the T level directly — restore Vit D + sleep architecture, expect T to rise to 600-800 by Day 90.
- `rdw_cv` 14.1 — Mild leaf of low-grade systemic perturbation. Will likely normalize as roots resolve. Monitor.
- `cortisol_am` 20 — Leaf of circadian misalignment, NOT chronic stress (PSS-4 = 4 disproves stress interpretation).
- `monocytes` 10.1% — Top of range — possibly post-Thailand/Bihar travel residue or just constitutional. Recheck CBC in 4-6 weeks.

- pdw 20.4 — Above range but isolated and clinically insignificant in absence of other platelet abnormalities. Monitor.

Biological age (Levine PhenoAge 2018)

Your biological age is 22.3 — 2.7 years younger than your calendar.

PhenoAge is a validated 9-marker blood-chemistry estimate of how your body is aging, trained on 11+ years of NHANES mortality data. It is not a diagnosis or a destiny — it's a snapshot, and it is movable. Your reading reflects a strong cardiometabolic-immune-renal-hepatic chassis. Even small gains from Vit D restoration + circadian fix could bring this to 21.5–22.0 by Day 90.

The 9 inputs used:	
albumin g/L	50
creatinine umol/L	90.17
glucose mmol/L	4.384
hs crp mg/L	0.9
lymphocytes pct	39.1
mcv fL	89
rdw cv pct	14.1
alp U/L	90
wbc K/uL	5.29
age	25
xb (linear combination)	-10.8344
Computed bio age	22.3 years

YOUR PANEL COVERAGE

Markers we'd add at Day 90 retest

Your current panel covers the metabolic + cardiometabolic + endocrine + hematology foundations well. To reach a complete Tessera longevity panel — and to interpret your borderline findings with higher confidence — consider these add-ons at retest. Ranked by clinical leverage.

MUST-ADD (WOULD CHANGE INTERPRETATION OF CURRENT FINDINGS)

- `free_t` — Completes T context — your total T 502 is below Tessera-optimal 600+; need Free T + SHBG to interpret bioavailable fraction
- `shbg` — Needed to compute bioavailable testosterone
- `anti_tpo` — Screens for autoimmune thyroid given borderline-high eosinophils 9.4% (Th2/atopic bias) — Hashimoto correlates with atopic phenotype
- `ige_specific_panel` — If atopic HPI confirmed (rhinitis/eczema/asthma) — total IgE 30 is normal but specific panel identifies triggers
- `vitamin_d_recheck_at_week_12` — Mandatory recheck post-loading dose

SHOULD-ADD (NEXT-LAYER SPECIFICITY)

- `dhea_s` — Adrenal reserve given cortisol AM 20 (high-normal); rules in/out HPA fatigue
- `igf_1` — Anabolic axis check, especially relevant for young M with optimization goals
- `cystatin_c` — Confirms kidney function; creatinine 1.02 sits at high-normal — could be muscle mass or early GFR drift

NICE-TO-ADD (RESEARCH-GRADE OPTIMIZATION)

- `glyca` — NMR chronic-inflammation marker; finer-grained than hsCRP for monitoring
- `omega_3_index` — Cellular EPA+DHA — strong CV mortality predictor, independent of lipids
- `hs_troponin` — Research-grade chronic myocardial stress; useful baseline at age 25

- nt_probnp — Atrial wall stress baseline

Full biomarker analysis

78 markers, grouped by panel

Status badges: **OPTIMAL** within Tessera-optimal range · **WATCH** within acceptable but not optimal · **OFF-TARGET** outside acceptable · **REFER** Tier-1 physician referral.

Complete Blood Count

Hemoglobin	15.8 g/dL	14–16	OPTIMAL
Red Blood Cells	5.25 10^6/uL	4.7–5.3	OPTIMAL
Hematocrit / PCV	46.8 %	42–48	OPTIMAL
Mean Corpuscular Volume	89 fL	85–92	OPTIMAL
MCH	30.2 pg	28–32	OPTIMAL
MCHC	33.9 g/dL	33–35	OPTIMAL
RDW-CV	14.1 %	≤ 13	WATCH
Total Leucocyte Count	5.29 10^3/uL	4.5–6.5	OPTIMAL
Neutrophils %	41 %	50–60	WATCH
Lymphocytes %	39.1 %	28–40	OPTIMAL
Monocytes %	10.1 %	3–7	OFF-TARGET
Eosinophils %	9.4 %	1–3	OFF-TARGET
Basophils %	0.4 %	0–1	OPTIMAL
Absolute Neutrophil Count	2.17 10^3/uL	—	OPTIMAL
Absolute Lymphocyte Count	2.07 10^3/uL	—	OPTIMAL
Absolute Monocyte Count	0.53 10^3/uL	—	OPTIMAL
Absolute Eosinophil Count	0.5 10^3/uL	—	OPTIMAL
Absolute Basophil Count	0.02 10^3/uL	—	OPTIMAL
Platelet Count	203 10^3/uL	200–350	OPTIMAL
Mean Platelet Volume	10.8 fL	—	OPTIMAL
Platelet Distribution Width	20.4 fL	—	OPTIMAL
ESR	3 mm/hr	—	OPTIMAL

Glucose & Insulin

Fasting Glucose	79 mg/dL	75-88	OPTIMAL
HbA1c	4.8 %	≤ 5.3	OPTIMAL
Fasting Insulin	3.8 uU/mL	2-6	OPTIMAL

Lipids

Total Cholesterol	146 mg/dL	150-200	WATCH
Triglycerides	56 mg/dL	≤ 90	OPTIMAL
HDL Cholesterol	49 mg/dL	≥ 45	OPTIMAL
LDL Cholesterol	86 mg/dL	≤ 100	OPTIMAL
Non-HDL Cholesterol	97 mg/dL	≤ 130	OPTIMAL
Lipoprotein(a)	11.4 mg/dL	≤ 30	OPTIMAL
Apolipoprotein A1	134 mg/dL	≥ 130	OPTIMAL
Apolipoprotein B	70 mg/dL	≤ 80	OPTIMAL
ApoB:ApoA1 Ratio	0.52 ratio	≤ 0.6	OPTIMAL

Derived Indices

HOMA-IR	0.741 index	≤ 1	OPTIMAL
TG:HDL Ratio	1.143 ratio	≤ 1.5	OPTIMAL
Atherogenic Index of Plasma	0.058 log10(TG/HDL)	≤ 0.11	OPTIMAL
Castelli Index I (TC:HDL)	2.98 ratio	≤ 3.5	OPTIMAL
Castelli Index II (LDL:HDL)	1.76 ratio	≤ 2.5	OPTIMAL
FIB-4 Liver Fibrosis Index	0.687 index	≤ 1.3	OPTIMAL
Neutrophil:Lymphocyte Ratio	1.05 ratio	1-2	OPTIMAL
Monocyte:HDL Ratio	0.0108 ratio	≤ 0.01	WATCH
Transferrin Saturation	36.5 %	25-35	WATCH

Inflammation

High-sensitivity CRP	0.9 mg/L	≤ 0.5	WATCH
C-Reactive Protein	1.1 mg/L	—	OPTIMAL
Homocysteine	7.46 umol/L	≤ 7	WATCH
Rheumatoid Factor	20 IU/mL	—	OPTIMAL

Liver

ALT (SGPT)	17 U/L	≤ 25	OPTIMAL
AST (SGOT)	23 U/L	≤ 25	OPTIMAL
Alkaline Phosphatase	90 U/L	50-100	OPTIMAL
GGT	11 U/L	≤ 30	OPTIMAL
Total Bilirubin	0.7 mg/dL	0.4-1	OPTIMAL
Total Protein	7.9 g/dL	6.6-8	OPTIMAL
Albumin	5 g/dL	4.3-5	OPTIMAL

Kidney & Electrolytes

Creatinine	1.02 mg/dL	0.8-1	WATCH
Uric Acid	3.7 mg/dL	3.5-5.5	OPTIMAL
Sodium	138 mmol/L	138-142	OPTIMAL
Potassium	4.3 mmol/L	4-4.5	OPTIMAL
eGFR	105 mL/min/1.73m2	≥ 90	OPTIMAL
Calcium	9.4 mg/dL	9.4-10	OPTIMAL
Phosphorus	4.6 mg/dL	2.8-4	OFF-TARGET

Thyroid

TSH	2.064 uIU/mL	1-2	WATCH
Total T3	0.97 ng/mL	—	OPTIMAL
Total T4	6.7 ug/dL	—	OPTIMAL
Free T3	2.84 pg/mL	3-3.8	WATCH
Free T4	1.01 ng/dL	1.1-1.4	WATCH

Vitamins & Iron

Vitamin D (25-OH)	15 ng/mL	40-60	REFER
Vitamin B12	508 pg/mL	500-900	OPTIMAL
Folate (B9)	8.1 ng/mL	≥ 10	WATCH
Serum Iron	104 ug/dL	80-130	OPTIMAL
TIBC	285 ug/dL	250-350	OPTIMAL
Ferritin	137.75 ng/mL	75-150	OPTIMAL

Hormones

Total Testosterone	502.08 ng/dL	600-900	WATCH
Cortisol AM	20 ug/dL	8-15	WATCH
Total PSA	0.481 ng/mL	≤ 1	OPTIMAL

Allergy

Total IgE	30 IU/mL	≤ 50	OPTIMAL
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Pancreas

Lipase	30 U/L	—	OPTIMAL
Amylase	47 U/L	—	OPTIMAL

Your protocol — 3 pillars, in order

What to do, why, what to expect

Lead with the highest-leverage root. Pillar 1 alone will move 3 of your 4 sub-optimal findings.

PILLAR 1

Pillar 1 — Vitamin D restoration (the single highest-leverage move)

Sits upstream of 3 of your 4 sub-optimal findings: atopy (Th2 bias), relative-low T (steroidogenic cofactor), borderline RDW (immune modulation). Move this one number and three others normalize.

ANCHOR MOVES

- Vitamin D3 60,000 IU/week × 8-12 weeks with fatty meal (physician supervised)
- K2 (MK-7) 100µg/day + Mg glycinate 300mg bedtime
- AM sunlight 15 min/day on forearms — Hyderabad before 10am or after 4pm
- Maintenance 2,000 IU/day after week-12 retest

Expected to shift: vitamin_d_25oh → 40-60 ng/mL, eosinophils ↓ (partial), testosterone → 600-800, RDW → <13.5, mood + sleep modulation

PILLAR 2

Pillar 2 — Circadian shift + sleep architecture (T-optimization lever)

Your AM cortisol 20 is not chronic stress (PSS-4 = 4) — it's a phase-shifted curve from late dinner (11pm-12am) + late bedtime (~1am). T peaks in deep-sleep windows. Compress these and T rises.

ANCHOR MOVES

- Move dinner to 8-9pm (drop 3 hours earlier from current 11pm-12am)
- Anchor bedtime by 11pm-midnight; anchor wake by 7-8am
- 15-min AM sunlight within 30 min of waking
- Phone out of bedroom; screen curfew 30 min before sleep (currently 8-9h daily screen)
- Magnesium glycinate 300mg at bedtime (also helps onset latency 20-40min)

Expected to shift: AM cortisol → 10-15 µg/dL, T → 600-800, sleep onset latency → <15 min, compressed deep-sleep window

PILLAR 3

Pillar 3 — Atopic + parasitic workup (parallel medical track)

Eosinophils 9.4% with confirmed seasonal rhinitis + recent travel to Bihar (rural India) + Thailand. You can't assume which it is — must test.

ANCHOR MOVES

- ****Medical referral****: Primary care for stool ova-and-parasite (O&P) panel — rule out helminths/giardia post-Bihar travel. Cheap and definitive.
- ****Medical referral****: Allergist consult if rhinitis severity rises above 6/10, OR if AEC stays elevated at Day-90 retest. Specific-IgE panel (dust mite, pollen, mold).
- Quercetin 500mg × 2/day + omega-3 2g/day (already in supplement stack)
- HEPA filter in bedroom — Hyderabad winter pollen + dust mite
- Shower before bed during high-pollen seasons (Aug-Sep, Nov-Feb)

Expected to shift: eosinophils → <4%, rhinitis severity → <2/10 with avoidance + Vit D

5-domain protocol

The details

NUTRITION

3 surgical shifts to your already-good diet

1. Your existing diet is solid (chicken + roti + curries + protein shake + low alcohol + low caffeine). 3 surgical shifts:
2. 1. MOVE DINNER 3 HOURS EARLIER. Currently 11pm-12am → target 8-9pm. This is the single biggest non-supplement lever for T + cortisol + sleep architecture.
3. 2. COMPRESS EATING WINDOW. Currently 13h (11am→12am) → target 10-11h (e.g., 11am→9pm). You stated 10h but actual is 13h — fix the math.
4. 3. ADD ZINC + FATTY-FISH FREQUENCY. Salmon/mackerel 2×/week + 30g pumpkin seeds/day. Both support T + Vit D.

DAILY TARGETS

- Protein: 130g (1.55 g/kg)
- Fiber: 35g
- Eating window: 10 hours (actual currently 13h)

CONCRETE MEAL PATTERN

- Breakfast (11am)**: continue protein shake; add 2 boiled eggs + 1 cup berries or banana — adds 15g protein + antioxidants
- Lunch (2pm)**: keep heavy — chicken + rotis + dal + 1 cup palak/methi (cooked, lemon-dressed) + curd. This is your biggest meal — good.
- Snack (5:30-6pm)**: 30g almonds/walnuts + 1 fruit. Add 30g pumpkin seeds 3-4×/wk for zinc.
- **Dinner (8:30pm — MOVED EARLIER)**: lighter — chicken or paneer (150g) + 2 rotis + 1 cup sabzi (palak/methi/lauki) + small bowl curd. Skip the protein shake at dinner — you've hit your daily protein already.
- Salmon / mackerel 2×/wk replacing chicken at lunch or dinner

EMPHASIZE

fatty fish (salmon, mackerel, sardines) 2×/wk, egg yolks daily (Vit D + choline), pumpkin seeds 30g/day (zinc), leafy greens (palak, methi, moringa) daily, turmeric + black pepper (anti-inflammatory), berries / pomegranate (polyphenols).

ATOPIC-OVERLAY CONSIDERATION

consider 4-week elimination trial if rhinitis worsens: dairy, gluten, common nuts (not all at once).

TRAINING

You're already strong here — add Zone 2 base + NEAT

Your training is already excellent. Two adjustments: more Zone 2 base, more NEAT outside the gym.

WEEKLY STRUCTURE TARGET

- Strength: 3-4 PPL sessions/week (keep)
- High-intensity: 1-2 HIIT/boxing (cap at 2)
- Zone 2: 120 min/wk (vs current ~30 min)
- Mobility: 10 min/day
- NEAT steps: target 8,000/day (vs current 3,500)

12-WEEK RAMP

- **Weeks 1-2:** Move dinner to 9pm. Add 1× weekly long Zone 2 (60 min). NEAT to 6k/day.
- **Weeks 3-6:** Zone 2 to 90 min/wk (split into 2-3 sessions). NEAT to 8k/day. Keep strength + HIIT.
- **Weeks 7-10:** Zone 2 to 120 min/wk. Add 1× VO2max session (4×4 min hard intervals). Keep PPL.
- **Weeks 11-13:** Hold all habits. Week 12: fast 12h, no caffeine 24h, no alcohol 72h before retest.

Why this matters: You have insulin sensitivity (HOMA-IR 0.74) without the Zone 2 base. Building Zone 2 mitochondrial density lowers RHR, improves HRV, supports T optimization, and adds healthspan years — for someone with 4 grandparents past 85, this is the big lifelong lever.

RECOVERY & SLEEP — THE HIGHEST-LEVERAGE SHIFT

Your sleep quality is good (7-8/10) — the issue is the CLOCK, not the architecture. Shift the whole schedule 2-3 hours earlier.

DAILY WIND-DOWN (TARGET SCHEDULE)

- 20:30: dinner finished (3h before bed)
- 21:00-22:00: screen curfew — no laptop/phone (use red-light glasses if reading on phone)
- 22:00: 5-10 min slow breathing (4-7-8 or physiological sigh) — anchors parasympathetic
- 22:30: Mg glycinate 300mg with water
- 23:00: bed
- 07:00: wake + 15 min AM sunlight within 30 min (Hyderabad balcony or short walk)

ENVIRONMENT FIXES

- Room temp: aim for 22-24°C (use AC or fan; currently 'moderate-warm' — temperature drives sleep architecture)
- Light: blackout curtains if not present
- Phone: out of bedroom (charge in another room)
- HEPA filter in bedroom (also helps atopic)

Expected: Sleep onset 20-40 min → <15 min; consistent 8h at correct phase; AM cortisol normalizes; T pulses optimized; mood + cognition compound benefits.

SUPPLEMENTS

7 items · ₹6,500/mo (within ₹7,500 budget)

WHAT	WHY	EVIDENCE	COST
Vitamin D3 60,000 IU/week × 8 wk → 2,000 IU/day after	Pillar 1 — root cause	[E1]	₹300/mo
Vitamin K2 (MK-7) 100 µg/day	Direct Ca to bone during high-dose D loading	[E2]	₹400/mo
Magnesium glycinate 300 mg	Vit D cofactor + sleep onset + RBC-Mg optimization	[E1]	₹500/mo
Omega-3 EPA+DHA 2 g combined EPA+DHA/day	Th2 modulation (atopic) + CV optimization + replaces missing omega-3 index marker	[E2]	₹1500/mo
Quercetin 500 mg × 2/day with meals	Mast-cell stabilization + Th2 modulation (atopic)	[E3]	₹800/mo
Whey protein (continuing) 2 scoops/day (~50g)	Muscle-building primary goal	[E1]	₹2500/mo
Creatine monohydrate (continuing) 5 g/day	Strength + cognition + cellular hydration; safe at your kidney function (eGFR 105)	[E1]	₹500/mo

SCHEDULE

- **Morning + breakfast:** Vit D3 60k (Sundays only), K2 100µg, Quercetin 500mg, Whey 1 scoop, Creatine 5g
- **Lunch:** Omega-3 2g, Quercetin 500mg
- **Post-workout:** Whey 1 scoop
- **Before bed:** Magnesium glycinate 300mg

SAFETY-CLEARED

Every supplement above was checked against your medications (none), conditions (none), allergens (none), and life stage. 7 items cleared, 1 with note (verify omega-3 is 3rd-party tested for heavy metals).

ITEMS INTENTIONALLY SKIPPED

- **Ashwagandha KSM-66 600mg/day:** Originally would have fired from R-HOR-03 (stress) BUT downgraded — PSS-4 = 4 (LOW), mood 7-8, no GAD-2 signal. Adding ashwagandha at low-stress baseline = unnecessary and adds adaptogen load. Revisit only if cortisol AM persists elevated AFTER circadian shift at retest, or if PSS-4 rises.
- **Methylated B-complex:** Folate 8.10 + B12 508 + homocysteine 7.46 are all in range. Mild RDW signal can be monitored. Skip unless RDW persists elevated at retest.

MIND & WELLNESS

Low-touch — your mental health profile is healthy

Your mental health profile is essentially healthy. PSS-4 low, mood 7-8, fatigue 1, brain fog 1, GAD-2 negative. PHQ-2 at threshold (3) is borderline — consider a brief check-in if it persists.

DAILY

5-10 min slow breathing (4-7-8 or physiological sigh) at bedtime — primary anchor for parasympathetic shift + sleep onset.

WEEKLY

1 outdoor walk in nature (Hyderabad has Necklace Road, Lumbini Park, KBR Park — get 30+ min weekly).

SCREEN TIME

Phone/laptop off 60 min before bed. Currently 8-9h screen time daily — keep the volume but cap the timing.

PHQ-2 FOLLOW-UP

If borderline PHQ-2 persists at next intake (90 days), book one session with a counselor — iCall India (9152987821, free) or a PCP. Not urgent now but worth tracking.

YOUR 90-DAY HABIT STACK

Six habits, four blocks

Cap of 6 habits total. More than this and adherence collapses. The order matters: foundation first, then layer.

WEEKS 1-2

- Anchor wake 7:00 daily incl. weekends
- 15-min AM sunlight within 30 min of waking
- Move dinner to 9:00pm

WEEKS 3-6

- + Vit D 60k Sunday + K2 + Mg + omega-3 + quercetin (full stack on)
- + Bedtime 23:00 anchor

WEEKS 7-10

- + Zone 2 to 120 min/wk (long run weekly + 2× easy cycle/treadmill)
- + NEAT to 8k steps/day

WEEKS 11-13

- Maintain all 6 habits + pre-retest prep (week 12: fast 12h, caffeine off 24h, alcohol off 72h)

RETEST PLAN

Per-marker cadence — not one-size-fits-all

DAY 30 — EARLY SIGNALS

- `cortisol_am` — expected ↓ 2-4 µg/dL. *Earliest signal of circadian shift working*
- `CBC differential (eos)` — expected eosinophils ↓ 1-2%. *Atopic-vit-D axis early signal; also confirms parasitic differential if O&P negative*

WEEK 12 — VIT D SPECIFIC

`vitamin_d_25oh` — expected 40-60 ng/mL (mandatory post-loading dose recheck).

DAY 90 — FULL PANEL + ADD-ONS

Repeat current panel + the must-add markers from the panel-completeness section above.

Estimated must-add cost: ₹3,500. Full add-on (incl. research-grade): ₹12,300.

WHAT WE EXPECT TO SEE AT DAY 90

Outcome expectations

MARKER	TODAY	DAY 90 TARGET
<code>vitamin_d_25oh</code>	15.0	40-60 ng/mL
<code>eosinophils_pct</code>	9.4	<4% (assuming O&P negative; if positive + treated, faster)
<code>testosterone_total</code>	502	600-800 ng/dL
<code>rdw_cv</code>	14.1	13.0-13.5%
<code>cortisol_am</code>	20.0	10-15 µg/dL (consistent ~8am draw)
<code>bio_age</code> (PhenoAge)	22.3	21-22 (marginal additional gain — already excellent)

Recommended physician referrals

Three medical actions

TIER 1 — Severe Vitamin D deficiency (15 ng/mL, threshold <20)

Primary care or Endocrinology · Sign-off on loading dose protocol (60,000 IU/wk × 8-12 wk); rule out malabsorption or hyperparathyroidism if no response at week 12; check ionized calcium baseline

Eosinophilia 9.4% + recent Bihar/Thailand travel — parasitic workup

Primary care · Stool ova-and-parasite (O&P) panel × 3 samples; serology for strongyloides, schistosoma (depending on travel itinerary); CBC differential repeat in 4-6 weeks

Seasonal rhinitis severity 4/10 with confirmed atopic phenotype

Allergist (only if symptoms persist past Vit D restoration at week 12) · Specific-IgE panel (dust mite, pollen, mold); pulmonary function test if any cough/breathing symptoms emerge

IMPORTANT NOTES

This report is generated from a single blood test + lifestyle intake. It is **not a medical diagnosis**. Every recommendation must be discussed with your physician before changing prescribed medications, starting new supplements at therapeutic doses, or modifying lifestyle in ways that affect existing health conditions.

The Tier-1 finding (Vitamin D 15 ng/mL) requires physician supervision of the loading dose. The eosinophilia + travel history requires stool ova-and-parasite testing through a physician. Do not start the supplement protocol until the Tier-1 review is complete.

Tessera is a biomarker-moving and healthspan-optimization program. We do not claim to extend lifespan, reverse aging, or cure disease. We work to move biomarkers in directions associated with better long-term health outcomes per current scientific evidence.

Privacy: this report and the underlying lab + intake data live in your secure run folder. They are not shared with employers, insurers, or third parties without your explicit consent (DPDP Act 2023 compliant).

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